Learning Objectives

- Understand integration as the reverse process of differentiation.
  - Solutions for indefinite integrals should include an arbitrary constant.
- Integrate sums of terms in powers of x , including $\frac{1}{x}$ and $\frac{1}{ax+b}$.
  - Solutions for indefinite integrals should include an arbitrary constant.
- Integrate functions of the form:
  - $(ax+b)^n$ for any rational n .
    - Includes the case where $n=-1$.
  - $\sin(ax+b)$
  - $\cos(ax+b)$
  - $\sec^2(ax+b)$
  - e^{ax+b}
  - For trigonometric functions angles will always be in radians.
- Evaluate definite integrals.

List of Terms

Arbitrary Constant 任意常数	An arbitrary constant is a constant value that appears in the general solution to an indefinite integral. Typically denoted as $\boldsymbol{c}$.
Integration 积分	Integration is a fundamental operation in calculus that involves finding the integral of a function. It is the inverse operation of differentiation.
Indefinite Integral 不定积分	The indefinite integral is a type of integral that does not have specified limits. The result of an indefinite integral includes an arbitrary constant, denoted as c.
Definite Integral 定积分	A definite integral calculates the accumulation of a quantity over a specific interval, defined by upper and lower limits. Unlike the indefinite integral, it does not include an arbitrary constant.

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Important Notes

Process of integration – Reverse process of differentiation	
$\frac{d}{dx}F(x) = f(x) \rightarrow \int f(x)\,dx = F(x) + c$	Where c is an arbitrary constant.
$\int [\alpha f(x) + \beta g(x)]dx = \alpha \int f(x)dx + \beta \int g(x)dx$	Where α and β are constants.

Indefinite integrals of some standard functions
Simple algebraic functions $$\int ax^n\,dx = \frac{ax^{n+1}}{n+1} + c, \qquad n \neq -1$$ $$\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c, \qquad n \neq -1, \qquad a \neq 0$$
Trigonometric functions $$\int \cos x\,dx = \sin x + c$$ $$\int \sin x\,dx = -\cos x + c$$ $$\int \sec^2 x\,dx = \tan x + c$$ **For** $a \neq 0$, $$\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + c$$ $$\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + c$$ $$\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + c$$
Exponential functions $$\int e^x\,dx = e^x + c$$ $$\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + c, \qquad a \neq 0$$

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Definite integrals
$\frac{d}{dx} F(x) = f(x) \rightarrow \int_a^b f(x)\, dx = F(b) - F(a)$
$\int_a^b [a\, f(x) + b\, g(x)]\, dx = a \int_a^b f(x)\, dx + b \int_a^b g(x)\, dx$
1) $\int_a^a f(x)\, dx = 0$ 2) $\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx$ 3) $\int_a^b f(x)\, dx + \int_b^c f(x)\, dx = \int_a^c f(x)\, dx$

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Warm Up

Follow along with the teacher to complete the following warm-up activities.

1. Find y if $\frac{dy}{dx}$ is given by

a) $3x^3$ **b)** $\frac{1}{x^2}$ **c)** $\sqrt{x}$

d) 5

2. Find y if $\frac{dy}{dx}$ is given by

a) $x^2 + 2$ **b)** $x(x + 3)$ **c)** $\frac{x^2-1}{x^2}$

d) $\frac{x^2+1}{\sqrt{x}}$

3. Find the equation of the curve which passes through the point (2, 3) and for which $\frac{dy}{dx} = 3x^2 + x$

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4. Find A as a function of x given that $\frac{dA}{dx} = x(x-1)$ and that $A = 5$ when $x = 3$.

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